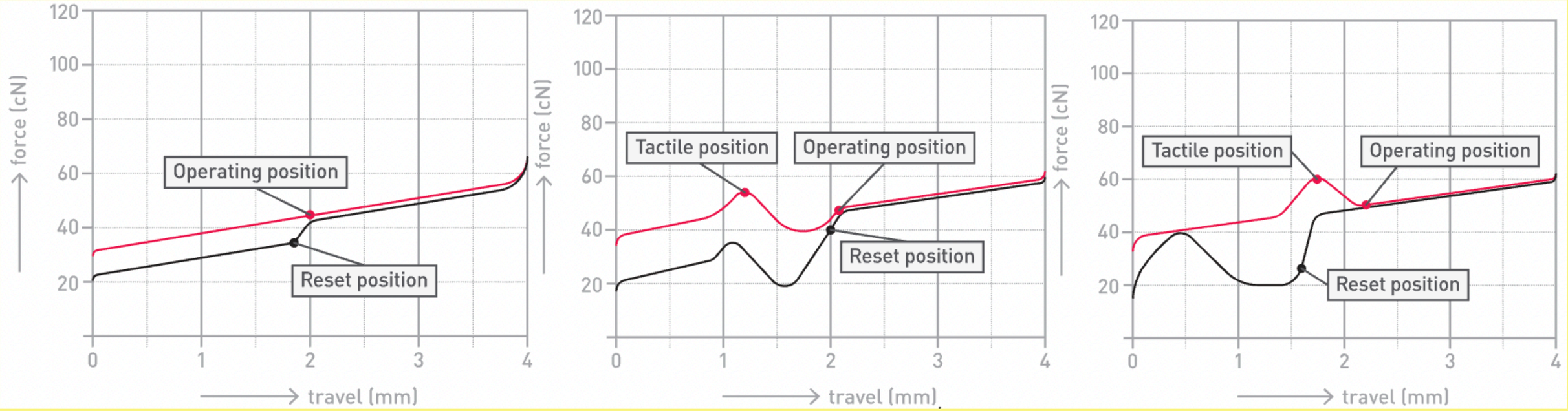
# Optical keyboard switches

**Corey Power De Jong - 18342643**
corey.powerdejong@ucdconnect.ie

# Mechanical switches

- Gold standard for keyboard switches
- Vague criteria, but generally agreed upon
- Discrete switches
- Stem pressed down causes metal contacts to touch
- Current flows, registering keypress
- Spring returns stem to unpressed position




Courtesy of Cherry MX switch datasheets

# Linear

# Tactile

# Clicky

# Optical switches

- Broadly similar to traditional mechanical switches
- Uses an infrared emitter and phototransistor
- Stem blocks infrared beam when pressed
- Keyboard detects the loss of light as a keypress
- Less wear and tear on metal contacts
- Fast response time

# Cost of switches

- Checked price per unit at 1000 units
- Phototransistor: 12c
- IR emitter: 8c
- At least 20c per switch before markup, housing moulds, manufacturing
- Current budget switches retail around 30c per switch

# Overview

- Not yet ready to overtake mechanical switches
- Expensive, not widely compatible with current enthusiast keyboards
- May take off in the future
- Currently only used for professional gaming keyboards

# References


https://www.cherrymx.de/_Resources/Persistent/b/7/a/b/b7ab2f72bd5686e1e1e759a9f3703536d7fb1e18/EN_CHERRY_MX_RED.pdf

https://www.cherrymx.de/_Resources/Persistent/1/3/6/1/13618248706cd28e75ab9bdf9e55e9f8794611c1/EN_CHERRY_MX_BROWN.pdf

https://www.cherrymx.de/_Resources/Persistent/a/5/3/1/a531cb6598bc849cbcf131fd7a31814282b74545/EN_CHERRY_MX_BLUE.pdf

https://www.razer.com/eu-en/razer-optical-switch

https://www.kingbrightusa.com/images/catalog/SPEC/APA3010P3BT-GX.pdf

https://www.digikey.ie/en/products/detail/kingbright/APA3010P3BT-GX/4977013

https://pdf1.alldatasheet.com/datasheet-pdf/view/455255/EVERLIGHT/IR12-21C-TR8.html

https://www.digikey.ie/en/products/detail/everlight-electronics-co-ltd/IR12-21C-TR8/2676085

# Thank you!

- Questions?